Parity of ranks in curve families

Problèmes ouverts — A. Dujella · 2026

Problem 2.5 (Dujella, 2026). Propose a heuristic to predict the distribution of parities of ranks within known families of elliptic curves over $\mathbb{Q}$ with the torsion group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$ (respectively $\mathbb{Z}/8\mathbb{Z}$) and rank $\geqslant 3$.

Remark. Certain experimental data concerning the parity of ranks in some of these families can be found in Dujella, A., Kazalicki, M., Peral, J.C. : Elliptic curves with torsion groups $\mathbb{Z}/8\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/6\mathbb{Z}$. Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Math. RACSAM **115**, Article 169 (2021).